

Sound is measured in decibels or dB.

Level and source of damaging noise include -

- 85-100 decibels - for example, hair dryer, blender power lawn mower, subway train etc.
- 100-120 decibels - for example - Bulldozer, noise of machine shop, impact wrench, motorcycle etc.
- 120-140 decibels - Such as rock concerts, auto racing or hammer pounding a nail.
- 125-155 decibels - Like firecrackers or firework or Jet engine
- 170-195 decibels - For example short gun, blast and rocket lift off etc.

2. Shrillness of sound is determined by

- (a) velocity of sound (b) amplitude of sound
(c) wavelength of sound (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

Shrillness of the sound is determined by the Frequency which depends upon the wavelength.

Velocity of the sound = wavelength × Frequency

In simpler terms, the higher the frequency of a sound wave, the shriller it will be perceived by our ears.

3. The velocity of sound in air is approximately-

- (a) 10 km/sec (b) 10 mile/min
(c) 330 m/sec (d) 3×10^{10} /sec

42nd B.P.S.C. (Pre) 1997

Ans. (c)

The speed of sound varies from nature to the medium. Sound travels slowest in gases, it travels faster in liquids and fastest in solids. For example - sound travels at 330 m/s in air, it travels at 1531 m/s in sea water and at 5950 m/s in iron.

4. When a sound wave goes from one medium to another, the quantity that remains unchanged is :

- (a) Frequency (b) Amplitude
(c) Wavelength (d) Speed
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

When a sound wave goes from one medium to another, its frequency usually remains the same because it is like a driven oscillation and maintains the frequency of the original source while its speed and wavelength are changed. The amplitude of sound is also changed as it decreases with distance from its source.

5. Before playing the orchestra in a musical concert, a sitarist tries to adjust the tension and pluck the string suitably. By doing so, he/she is adjusting

- (a) frequency of the sitar string with the frequency of other musical instruments
(b) amplitude of sound
(c) intensity of sound
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Before playing the orchestra in a musical concert, a sitarist tries to adjust the tension and pluck the strings suitably. By doing so he is adjusting frequency of the sitar string with the frequency of other musical instruments. This process is known as tuning the sitar.

The pitch of a musical note is directly related to its frequency—the higher the frequency, the higher the pitch and vice versa.

Electric Current

1. If a bulb of 100 watt burns for 10 hours, the expenditure of electricity will be –

- (a) 0-1 unit (b) 1 unit
(c) 10 units (d) 100 units

41st B.P.S.C. (Pre) 1996

Ans. (b)

Energy Consumption of 100 watt electric bulb which is used for 10 hours = 100×10
= 1000 watt hour
= 1 kilowatt hour
= 1 unit.

2. Electricity consumption bill is based on the measurement of –

- 1. Watt 2. Voltage**
3. Ohm 4. Ampere

Select your answer of the following codes -

- (a) Only 1 (b) 1 and 2
(c) 2 and 3 (d) 1 and 4

40th B.P.S.C. (Pre) 1995

Ans. (a)

Electricity consumption bill is based on the measurement of kilowatt hour. The kilowatt hour is the unit of electricity consumption, equal to 1000 watt in 1 hour (1 unit).

3. The current capacity of a fuse wire does not depend

- (a) on the length of wire
- (b) on the circumference of wire
- (c) on the material of wire
- (d) on the current flowing through it
- (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (e)

The current rating is the maximum current that a fuse will carry for an indefinite period without too much deterioration of the fuse element. The current carrying capacity of a fuse wire depends on the following factor :

- Cross section area
- Dimension
- Type of material
- Shape of the terminal
- Temperature

4. The power in the supply line is transferred at high voltage

- (a) to transfer more power
- (b) to reduce power loss
- (c) to flow more current
- (d) to minimize power loss
- (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (c)

The primary reason that power is transmitted at high voltages is to increase efficiency. As electricity is transmitted over long distances, there are inherent energy losses along the way. High voltage transmission minimizes the amount of power lost as electricity flows from one location to the next.

5. The total resistance of a circuit having two parallel resistors is 1.403 kilo-ohm. If one of the resistors is 2.0 kilo-ohm, then the other resistor will be

- (a) 1.403 kilo-ohm
- (b) 2.0 kilo-ohm
- (c) 3.403 kilo-ohm
- (d) 4.70 kilo-ohm
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

If total resistance of a parallel circuit is R, then

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

According to the question,

$$\begin{aligned} \frac{1}{1.403} &= \frac{1}{2} + \frac{1}{R_2} \\ \text{or, } \frac{1}{R_2} &= \frac{1}{1.403} - \frac{1}{2} \\ \text{or, } \frac{1}{R_2} &= \frac{2 - 1.403}{2.806} \\ \text{or, } R_2 &= \frac{2.806}{0.597} = 4.70 \text{ kilo-ohm} \end{aligned}$$

6. The point to be kept in mind for verification of Ohm's law is

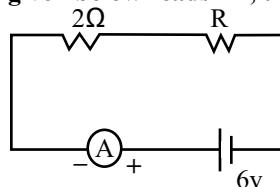
- (a) ammeter and voltmeter should be connected in series
- (b) ammeter should be connected in series and voltmeter in parallel
- (c) ammeter should be connected in parallel and voltmeter in series
- (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The point to be kept in mind for verification of ohm's law is ammeter should be connected in series because current remains same in series connection and voltmeter in parallel because potential difference in parallel connection remains same. Hence option (b) is the correct answer.

7. If the ammeter in the circuit shown in the diagram given below reads 2A, the resistance R is

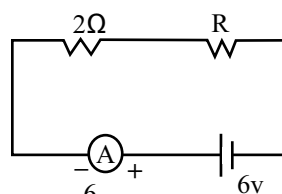


- (a) 1Ω
- (b) 2Ω
- (c) 3Ω
- (d) 4Ω
- (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (a)

In the given circuit diagram the Ammeter reads 2A.



$$\begin{aligned} I &= \frac{6}{2 + R} \\ 2 &= \frac{6}{2 + R} \\ 4 + 2R &= 6 \\ 2R &= 2 \\ R &= 1\Omega \end{aligned}$$

8. 'Dynamo' is a device, which converts :

- (a) Chemical energy into electrical energy
- (b) Electrical energy into mechanical energy
- (c) Mechanical energy into electrical energy
- (d) Electrical energy into chemical energy

39th B.P.S.C. (Pre) 1994

Ans. (c)

9. Turbines and Dynamos are used to convert which energy to electrical energy?

- 44th B.P.S.C. (Pre) 2000**

See the explanation of the above question.

47th B.P.S.C. (Pre) 2005

When electrical energy is converted into motion, there is no heat loss.

64th B.P.S.C. (Pre) 2018

Electric motor is a device which converts electrical energy into mechanical energy. Electric motors involve rotating coils of wire which are driven by the magnetic force exerted by a magnetic field or an electric current.

B.P.S.C. Auditor 2021

Every magnet has the capacity to sustain the magnetic property. When we start heating the magnet, it starts losing its magnetic property. If the magnet is strongly heated then it cannot hold its magnetic property under that high temperature. Thus, it gets demagnetized losing all of its magnetic properties..

The curie temperature is a temperature at which certain magnetic materials undergo a sharp change in their magnetic properties.

B.P.S.C. CDPO 2018

A fuel cell is a device that generates electricity through an electrochemical reaction, not combustion. In a fuel cell, hydrogen and oxygen are combined to generate electricity.

66th B.P.S.C. (Pre) 2020

The Faraday constant represents the magnitude of electric charge per mole of electrons. It is denoted by the symbol 'F' and is named after English scientist Michael Faraday. It is a universal constant. It has the currently accepted value $F = 96485.332123 \text{ C.mol}^{-1}$.

44th B.P.S.C. (Pre) 2000

Tungsten metal is used as filament in lighting bulbs. In 1781, Carl Wilhelm Scheele and Torbern Bergman suggested that it might be possible to obtain a new metal by reducing tungsten acid. Two years later in 1783, Juan Jose and Fausto Elhuyar were the first to isolate tungsten through the reduction of this acid with charcoal. For this reason, they are credited with the discovery of this element.

43rd B.P.S.C. (Pre) 1999

Ans. (c)